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How LLMs store facts II

1) MLPs ^{context} → MLP blocks are where models bake in general Knowledge and Facts

^{Scalar} → hold vast majority of model's Params

^{Indirection} → process each Token's vec in parallel

Architecture : → consists of pair of matrix multiplications with non-linear func in between.

Vec- E → mult by w_{up} → ReLU → mult by w_{down} → Add to E

Transformers :-

→ Before Transformers RNN were used for language

→ Problem : → They forget early parts of long paragraphs

→ can't be trained in parallel because they were sequential

→ Sol : → Transformers : process all words at once

→ scale efficient

→ can be trained on massive datasets

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RNN and Their issues:-

→ used for sequential data

Types {
→ many to one → Sentiment Analysis
→ many to many → Language Translation

Issues

→ During backpropagation chain rule multiplies many small or large numbers → causes gradient to shrink to zero or grow to infinity

LSTMs

→ Special Type of RNN

→ used to fix vanishing issues

BERT

→ Bidirectional Encoder Representations from Transformers

→ Reads ~~sequence~~ entire sequence at once

→ uses Transformer encoders

TF serving

→ High Performance System to deploy ML models

→ exposes model as REST API endpoint

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Denotational Semantics (old way)

- misses nuance ; doesn't keep up with slang
- Doesn't measure partial similarity well

Distributional Semantics (modern way)

- word defined by the contexts in which frequent^(*) appears

Word2Vec Algorithm

Goal: Learn Dense Vectors by predicting context of words

Process:

1) Start with random dense

2) Slide window through massive corpus of text

3) Take center word and use its vect to predict context words

4) calc dot product then use softmax to turn scores into probability

5) minimize loss during calculating to adjust vectors